

Aldrey Canlas

Senior BIM & AI Software Engineer

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SUMMARY

Senior BIM & Revit API Engineer with 10+ years of experience building productiongrade automation tools for architectural, engineering, and construction workflows. Specialized in developing custom Revit plugins using C# and the Revit API to automate modeling, data extraction, and cross-platform BIM interoperability (IFC, DWG, JSON). Proven background integrating generative AI, computational design, and cloud services into BIM environments to reduce manual effort, improve design accuracy, and accelerate delivery timelines. Strong collaborator with architects, BIM managers, and product teams, translating real-world design constraints into scalable, maintainable software solutions.

SKILLS

Core Competencies

- ✓ **BIM & Design Automation:** Revit API plugin development, BIM workflow automation, parameter management, standards enforcement, IFC/DWG interoperability
- ✓ **AI-Assisted Design Systems:** Generative design, AI-driven layout optimization, LLM-assisted tooling, design intelligence workflows
- ✓ **Applied Machine Learning:** Model integration, fine-tuning, inference pipelines, embeddings, multimodal AI systems
- ✓ **Software Architecture:** Scalable system design, modular plugin architectures, design patterns, maintainable codebases
- ✓ **Product & Tooling Development:** Interactive editors, internal tools, developer platforms, user-centric technical solutions
- ✓ **Engineering Leadership & Collaboration:** Cross-functional collaboration, technical mentorship, agile delivery, requirements translation

Technical Expertise

- ✓ **Programming:** Python, C#, JavaScript, TypeScript, HTML, CSS
- ✓ **BIM & CAD Technologies:** Revit API, BIM automation, IFC, DWG, computational design workflows
- ✓ **AI / ML Frameworks & APIs:** PyTorch, TensorFlow, Hugging Face Transformers, LangChain, OpenAI API, Gemini, SAM3, ControlNet
- ✓ **Frontend & UI Development:** React.js, React Native, interactive UI components, state-driven interfaces
- ✓ **DevOps & Infrastructure:** AWS, CI/CD pipelines, Docker (basic), Git, cloud-native deployment workflows
- ✓ **Testing & Quality:** Unit testing, integration testing, performance validation
- ✓ **Project & Delivery Tools:** Jira, Agile/Scrum workflows, version control best practices
- ✓ **AI-First Developer Tooling:** Replit, Lovable, AI-assisted development environments

PROFESSIONAL EXPERIENCE

Senior BIM & AI Software Engineer

Cognizant, United States (Remote)

Oct 2021 - Present

- Designed and developed advanced Revit API plugins in C# to automate BIM workflows for residential and commercial projects, including model generation, parameter management, and standards enforcement, significantly reducing manual modeling effort and improving cross-project consistency.
- Built BIM-integrated web applications using JavaScript, TypeScript, Angular, Node.js, C#, and AWS to support data-driven workflows, external model data consumption, and system interoperability.
- Created responsive web & mobile application for Visibly to perform thorough eye test.
- Provided technical guidance and mentorship on software engineering best practices, including Revit API development, BIM data structures, and scalable system design.

Technical Team Lead

Accenture, United States (Remote)

Oct 2019 - Oct 2021

- Implemented Revit API-based BIM systems to programmatically generate, modify, and manage families, parameters, and assemblies, enforcing design rules, validating model data, and adapting to evolving architectural and engineering constraints.
- Defined software architecture standards and development guidelines, making technical decisions across BIM automation tools, Revit plugins, and web-based systems to ensure long-term maintainability and scalability.
- Developed an advanced health monitoring and fitness coaching app, using Node.js and FastAPI for high-performance data synchronization between mobile devices and IoT sensors.
- Built internal desktop utilities with PyQt to automate test workflows and visualize user analytics data.
- Defined software development guidelines, made architectural decisions.

Software Engineer

Microsoft, United States (Remote)

Oct 2015 - Oct 2019

- Implemented scalable distributed Microservice in .Net Core using C# and leveraged Azure Traffic Manager for load balancing across geographies.
- Worked on a very large set of data to handle more than 1M daily records by utilizing asynchronous programming & robust queue data structure.
- Developed a SaaS platform for automated architectural design, combining BIM automation with AI-assisted generation for buildings and interior layouts.
- Worked on an internal Web application for analyzing and summarizing user search behavior and results.
- Created a Companion App, an app to provide insights about smartphones when connected via USB to Windows desktop.

Software Engineering Intern

Amazon, United States (Remote)

Jun 2015 - Oct 2015

- Designed and built back-end web service and front-end web page intended for use by fulfillment center operators to manage their associate's labor capabilities.
- Built back-end service using Amazon internal frameworks backed by Oracle SQL database.
- Built front-end web page using Spring MVC, Twitter bootstrap and Amazon internal frameworks.
- Worked on creating web widget to display keywords on video game landing pages.

EDUCATION

University of Santo Tomas
Bachelor's Degree in Computer Science

Philippines
2011 - 2015